Bureau
International des
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Estimates of the Differences between Thermodynamic Temperature the ITS-90

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Abstract

At the CCT's request, Working Group 4 (WG4) critically reviewed all available measurements of $T - T_{90}$ including constant-volume gas thermometry, acoustic gas thermometry, spectral radiation thermometry, total radiation thermometry, noise thermometry, and dielectric-constant gas thermometry. Consensus estimates are provided for $T - T_{90}$, for selected measurements from $4.2\ \mathrm{K}$ to $1\,358\ \mathrm{K}$, as well as a recommendation for analytic approximations to $T - T_{90}$ for the range $0.65\ \mathrm{K}$ to $1\,358\ \mathrm{K}$.

Estimates of the Differences between Thermodynamic Temperature and the ITS-90

Contents

1. Table of Differences 7
2. Interpolation Functions 8

References 10

1. Table of Differences

Table 1 summarizes the best estimates of $T - T_{90}$ above 4.2 K as of 2010. In general, a weighted average was formed using the uncertainties identified by WG4. For details see [1]. The data are shown in Figure 1 and Figure 2.

Table 1. Estimates of $T - T_{90}$ between 4.2 K and 1 358 K. The transitions of the defining fixed pointsand secondary reference points of the ITS-90 are marked in the 2nd and 6th columns. All uncertainties are standard uncertainties ($k = 1$). The differences for temperatures above 1 358 K are under investigation by Working Group 5. The results presented here may be extrapolated above 1 358 K using Planck's law.

T_{90} (K)		$T - T_{90}$ (mK)	u (mK)	T_{90} (K)		$T - T_{90}$ (mK)	u (mK)
4.2		–0.02	0.12	161.405	Xe	–8.43	1.8
5		0.10	0.12	195		–6.97	1.8
6		0.04	0.13	234.315 6	Hg	–3.25	1.0
7		–0.08	0.09	255		–1.64	0.9
8		0.01	0.10	273.16	TPW	0	0
9.288	Nb	0.13	0.11	290		2.19	0.4
11		0.27	0.12	302.914 6	Ga	4.38	0.4
13.803 3	e-H_2	0.44	0.14	335		7.62	0.5
17.035	e-H_2	0.51	0.16	373.124	H_2O	9.74	0.6
20.27	e-H_2	0.32	0.17	429.748 5	In	10.1	0.8
22.5		0.10	0.18	505.078	Sn	11.5	1.3
24.556 1	Ne	–0.23	0.20	600.612	Pb	9.21	6.1
35		–0.53	1.0	692.677	Zn	13.8	6.9
45		–0.75	1.4	800		22.4	6.4
54.358 4	O_2	–1.06	1.6	903.778	Sb	27.6	7.6
70		–1.57	1.9	933.473	Al	28.7	6.6
77.657		–3.80	1.2	1 052.78	Cu/Ag	40.9	26
83.805 8	Ar	–4.38	1.3	1 150		46.3	20
90		–5.30	1.1	1 234.93	Ag	46.2	14
100		–6.19	1.2	1 337.33	Au	39.9	20
130		–8.07	1.6	1 357.77	Cu	52.1	20

2. Interpolation Functions

If it is not convenient to use Table 1, the differences $T - T_{90}$ may be approximated by the following expressions. Above $70\ \mathrm{K}$, the relative differences of the interpolation functions (with respect to the values of Table 1, see p. 7) are less than 15 %, except at $600\ \mathrm{K}$ and the gold point.

From $0.65\ \mathrm{K}$ to $2\ \mathrm{K}$, use the polynomial for the temperature scale PTB-2006 (based on the ^{3}Helium vapor-pressure) [2] with

$$T - T_{90} \equiv T_{2\ 006} - T_{90}.$$

Below $1\ \mathrm{K}$, $T_{2\ 006}$ is identical to $T_{\mathrm{PLTS\text{-}2\ 000}}$.

From $2\ \mathrm{K}$ to $8\ \mathrm{K}$,

$$T - T_{90} \equiv 0.$$

From $8\ \mathrm{K}$ to $273.16\ \mathrm{K}$,

$$(T - T_{90})/\mathrm{mK} = \sum_{i=0..7} b_i \times \left(\log_{10}(T_{90}/273.16\ \mathrm{K})\right)^{i+1} \tag{1}$$

with the coefficients:

$$b_0 = 4.424\,57 \times 10^1 \quad b_1 = -1.763\,11 \times 10^2 \quad b_2 = -1.539\,85 \times 10^3 \quad b_3 = -3.636\,85 \times 10^3$$

$$b_4 = -4.198\,98 \times 10^3 \quad b_5 = -2.613\,19 \times 10^3 \quad b_6 = -8.419\,22 \times 10^2 \quad b_7 = -1.103\,22 \times 10^2$$

The derivative $\mathrm{d}(T - T_{90})/\mathrm{d}T_{90}$ at the triple point of water is 7.0×10^{-5}.

From $273.16\ \mathrm{K}$ to $1\,357.77\ \mathrm{K}$ (copper point):

$$(T - T_{90})/\mathrm{mK} = (T_{90}/\mathrm{K}) \sum_{i=0..4} c_i \times (273.16\ \mathrm{K}/T_{90})^{2i} \tag{2}$$

with the coefficients:

$$c_0 = 0.049\,7 \quad c_1 = -0.303\,2 \quad c_2 = 1.025\,4 \quad c_3 = -1.289\,5 \quad c_4 = 0.517\,6$$

The derivative at the triple point of water is 10.1×10^{-5}, resulting in a discontinuity of 3.1×10^{-5} between Equation (1) and Equation (2), see Figure 1. This is consistent with the values from recent thermodynamic measurements and measurements of platinum resistance thermometers that conform to ITS-90.

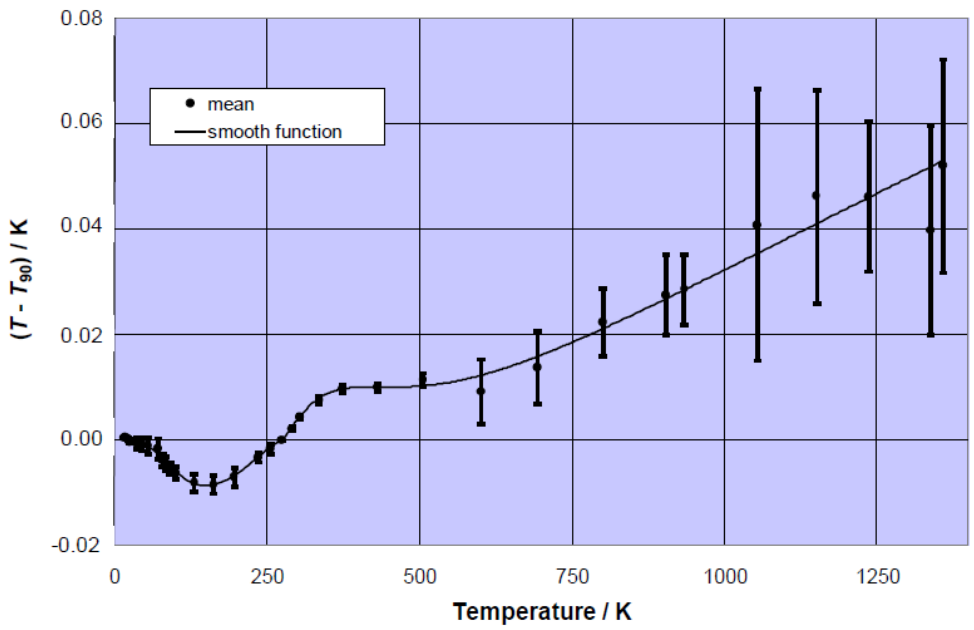

Figure 1 — Overview of consensus estimates for $T - T_{90}$ with emphasis on the range above the triple point of water. The smooth functions (Equation (1) and Equation (2), black line) are interpolating the mean values (black dots). Error bars represent uncertainties with $k = 1$.

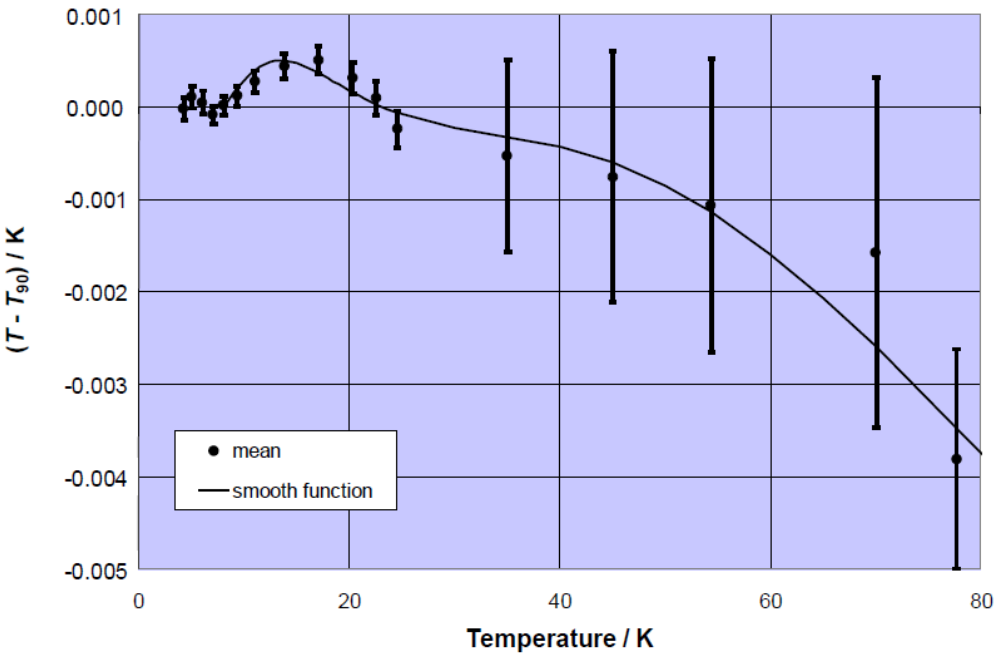

Figure 2 — Enlargement of the range between $4.2\ \mathrm{K}$ and $80\ \mathrm{K}$ of consensus estimates for $T - T_{90}$. The smooth function (Equation (1), black line) interpolates the mean values (black dots) above $8\ \mathrm{K}$. Error bars represent uncertainties with $k = 1$.

References

[1]] J. Fischer, M. de Podesta, K. D. Hill, M. Moldover, L. Pitre, R. Rusby, P. Steur, O. Tamura, R. White, L.Wolber, *Int. J. Thermophys.* **32**, 12-25 (2011).

[2]] J. Engert, B. Fellmuth, K. Jousten, *Metrologia* **44**, 40-52 (2007).

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